

Name: _____ Student ID: _____

ECON 308 – International Economic Relations

Final Exam

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Instructions:

- You are allowed **120 minutes** to complete the exam.
 - Show all work. Unsupported answers receive little credit.
 - Unless stated otherwise, all variables are real and differentiable as needed.
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Multiple Choice Answers:

- | | | | |
|-----------|-----------|-----------|-----------|
| 1. _____ | 15. _____ | 29. _____ | 43. _____ |
| 2. _____ | 16. _____ | 30. _____ | 44. _____ |
| 3. _____ | 17. _____ | 31. _____ | 45. _____ |
| 4. _____ | 18. _____ | 32. _____ | |
| 5. _____ | 19. _____ | 33. _____ | |
| 6. _____ | 20. _____ | 34. _____ | |
| 7. _____ | 21. _____ | 35. _____ | |
| 8. _____ | 22. _____ | 36. _____ | |
| 9. _____ | 23. _____ | 37. _____ | |
| 10. _____ | 24. _____ | 38. _____ | |
| 11. _____ | 25. _____ | 39. _____ | |
| 12. _____ | 26. _____ | 40. _____ | |
| 13. _____ | 27. _____ | 41. _____ | |
| 14. _____ | 28. _____ | 42. _____ | |

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question (1 point each, total of 45 points).

1. The slope of a country's production possibility frontier with cloth measured on the horizontal and food measured on the vertical axis in the Ricardian model is equal to ____ and it ____ as more cloth is produced.
 - A. $-MPL_F/MPL_C$; becomes flatter
 - B. $-MPL_F/MPL_C$; becomes steeper
 - C. $-MPL_F/MPL_C$; is constant
 - D. $-MPL_C/MPL_F$; is constant
 - E. $-MPL_C/MPL_F$; becomes steeper
2. Economists consider the effects of free trade on income distribution to be ____ important than the effects on overall welfare because ____.
 - A. less; those who are harmed can be compensated by those who gain
 - B. more; the effects on income distribution are major and consequential
 - C. more; those who are harmed are not compensated by those who gain
 - D. less; the effects on income distribution are minor and inconsequential
 - E. less; the wealthy benefit and only the poor lose
3. In the specific factors model, which of the following will increase the quantity of labor used in cloth production?
 - A. a decrease in the price of labor
 - B. an increase in the price of food relative to that of cloth
 - C. an increase in the price of cloth relative to that of food
 - D. an equal percentage increase in the price of food and cloth
 - E. an equal percentage decrease in the price of food and cloth
4. In the specific factors model, labor is defined as a(an)
 - A. intensive factor.
 - B. mobile factor.
 - C. specific factor.
 - D. fixed factor.
 - E. variable factor.

5. The specific factors model assumes that there are ____ goods and ____ factor(s) of production.
- A. four; three
 - B. three; two
 - C. two; one
 - D. two; two
 - E. two; three
6. In the specific factors model, the effects of trade on welfare are ____ for mobile factors, ____ for fixed factors used to produce the exported good, and ____ for fixed factors used to produce the imported good.
- A. positive; ambiguous; ambiguous
 - B. ambiguous; positive; negative
 - C. positive; positive; positive
 - D. ambiguous; negative; positive
 - E. negative; ambiguous; ambiguous
7. Factors tend to be specific to certain uses and products
- A. in the short run.
 - B. in capital-intensive industries.
 - C. in labor-intensive industries.
 - D. in countries lacking comparative advantage.
 - E. in countries lacking fair labor laws.
8. According to the Heckscher–Ohlin model, the source of comparative advantage is a country's
- A. technology.
 - B. factor endowments.
 - C. human capital.
 - D. advertising.
 - E. political system.

9. According to the Heckscher–Ohlin model
- A. everyone gains from trade.
 - B. a country gains from trade if its exports have a high value added.
 - C. the gainers from trade could compensate the losers and still retain gains.
 - D. only the country with the more advanced technology gains from trade.
 - E. the scarce factor gains from trade and the abundant factor loses.
10. Biased expansion of production possibilities occurs
- A. when the production possibility frontier shifts outward equally in the production of all goods.
 - B. when the production possibility frontier shifts inward more in one direction than in the other.
 - C. when the production possibility frontier shifts inward equally in the production of all goods.
 - D. when the production possibility frontier shifts outward much more in one direction than in the other.
11. Suppose that there are two factors, capital and land, and that the United States is relatively land-endowed while the European Union is relatively capital-endowed. According to the Heckscher–Ohlin model
- A. all landowners should support free trade.
 - B. all capitalists in both countries should support free trade.
 - C. the U.S. should compensate European countries once trade commences.
 - D. European capitalists should support U.S.–European free trade.
 - E. European landowners should support U.S.–European free trade.

12. Which of the following is an assertion of the Heckscher–Ohlin model?
- A. Factor endowments determine the technology that is available to a country, which determines the good in which the country will have a comparative advantage.
 - B. An increase in a country's labor supply will increase production of both the capital-intensive and the labor-intensive good.
 - C. In the long run, labor is mobile and capital is not.
 - D. Factor price equalization will occur only if there is costless mobility of all factors across borders.
 - E. An increase in a country's labor supply will increase production of the labor-intensive good and decrease production of the capital-intensive good.
13. An increase in the real interest rate, all other things held constant, will cause a country's ____ to ____.
- A. terms of trade; improve
 - B. terms of trade; worsen
 - C. welfare level; improve
 - D. current consumption; increase
 - E. current consumption; decrease
14. An increase in a country's net commodity terms of trade will
- A. not always guarantee positive changes in the country's economy.
 - B. always increase the country's real income.
 - C. always increase the country's production of its import-competing good.
 - D. always increase the country's economic welfare.
 - E. never increase the country's quantity of exports.
15. The distinctive feature of tariffs and export subsidies is that
- A. they make imported and exported goods cheaper.
 - B. they give incentives for importers and exporters.
 - C. they create a difference between prices at which goods are traded on the world market and prices at which those goods can be purchased within a country.
 - D. they equalize prices at which goods are traded on the world market and prices at which those goods can be purchased within a country.

16. A rise in the relative price of cloth leads the economy to produce ____ cloth and ____ food, therefore ____ the relative supply of cloth.
- A. more; less; decrease
 - B. more; less; increase
 - C. less; more; increase
 - D. less; more; decrease
17. The effect of an export subsidy is to
- A. give producers an incentive to export.
 - B. make it less profitable to sell abroad than at home.
 - C. raise government's revenue.
 - D. raise the prices of exported goods outside a country.
18. If points A and B are two locations on a country's production possibility frontier, then
- A. consumers are indifferent between the two bundles.
 - B. the country could produce either of the two bundles.
 - C. both bundles must have the same relative cost.
 - D. at any point in time, the country could produce both.
 - E. producers are indifferent between the two bundles.
19. If the ratio of the price of cloth (P_C) divided by the price of food (P_F) increases in the international marketplace, then
- A. the terms of trade of all countries will improve.
 - B. the terms of trade of cloth exporters will worsen.
 - C. the terms of trade of cloth exporters will improve.
 - D. all countries would be better off.
 - E. the terms of trade of food exporters will improve.

20. Patterns of interregional trade are primarily determined by ____ rather than ____ because factors of production are generally ____.
- A. external economies; population; immobile
 - B. population; external economies; immobile
 - C. internal economies; population; immobile
 - D. external economies; natural resources; mobile
 - E. internal economies; external economies; mobile
21. If a firm's output more than doubles when all inputs are doubled, production is said to occur under conditions of
- A. decreasing returns to scale.
 - B. constant returns to scale.
 - C. increasing returns to scale.
 - D. imperfect competition.
 - E. intra-industry equilibrium.
22. If some industries exhibit internal increasing returns to scale in each country, we should not expect to see
- A. intra-industry trade between countries.
 - B. inter-industry trade between countries.
 - C. increased productivity in both countries.
 - D. high levels of specialization in both countries.
 - E. perfect competition in these industries.
23. External economies of scale often arise because similar firms
- A. agree to cooperate to expand global trade.
 - B. have excellent internal logistics.
 - C. have economies of scale in production.
 - D. locate in the same geographic region.
 - E. collude to fix prices and increase profits.

24. External economies of scale will ____ cost per unit when output is ____ by ____.
- A. increase; increased; the industry
 - B. reduce; reduce; the industry
 - C. reduce; increased; a firm
 - D. reduce; increased; the industry
 - E. increase; increased; a firm
25. If output is increased in the long run, average production costs in the presence of internal economies of scale will ____, and in the presence of external economies of scale, will ____.
- A. increase; decrease
 - B. decrease; remain constant
 - C. decrease; decrease
 - D. increase; remain constant
 - E. remain constant; increase
26. The imposition of tariffs on imports results in deadweight (triangle) losses. These are
- A. production and consumption distortion effects.
 - B. distortion of incentives.
 - C. revenue effects.
 - D. efficiency effects.
 - E. redistribution effects.
27. The principle benefit of tariff protection goes to
- A. foreign consumers of the good produced.
 - B. domestic producers of the good produced.
 - C. foreign producers of the good produced.
 - D. domestic consumers of the good produced.
 - E. the domestic government.

28. An export subsidy ____ prices in the exporting country while ____ them in the importing country.
- A. lowers; lowering
 - B. lowers; raising
 - C. raises; lowering
 - D. raises; raising
29. Tariff rates on products imported into the U.S.
- A. were the government's main source of income in 2006.
 - B. reached an all time high in 2002.
 - C. were prohibited by the Constitution.
 - D. have dropped substantially over the past 50 years.
 - E. have risen steadily since 1920.
30. Trade between British Columbia and a Canada province is likely much larger than trade with an equally distant U.S. state because of
- A. size of local economy.
 - B. religion.
 - C. population.
 - D. negative effects of the borders.
31. A situation where a final product goes through many production stages by different companies in different countries instead of being produced by one company in one country is called
- A. horizontal integration.
 - B. vertical disintegration.
 - C. vertical integration.
 - D. horizontal disintegration.

- 32.** Since the early 1970s, world's trade as a share of world production has
- A. increased slightly before dropping off.
 - B. increased.
 - C. decreased.
 - D. remained constant.
 - E. fluctuated widely with no clear trend.
- 33.** In the present, most of the exports from China are
- A. technology-intensive products.
 - B. services.
 - C. primary products including agricultural.
 - D. manufactured goods.
 - E. overpriced by world market standards.
- 34.** In general, which of the following do *not* tend to increase trade between two countries?
- A. larger economies
 - B. linguistic and/or cultural affinity
 - C. mutual membership in preferential trade agreements
 - D. historical ties
 - E. the existence of well controlled borders between countries
- 35.** Which of the following statements is TRUE?
- A. The competitive advantage of an industry depends not only on its productivity relative to the foreign industry, but also on the domestic wage rate relative to the foreign wage rate.
 - B. The competitive advantage of an industry only depends on the domestic wage rate relative to the foreign wage rate.
 - C. The competitive advantage of an industry depends on a country's absolute productivity advantage over other countries.
 - D. The competitive advantage of an industry only depends on its productivity relative to the foreign industry.

36. When one country can produce a unit of a good with less labor than another country, the first country has a(n) ____ in producing that good.
- A. advanced advantage
 - B. comparative advantage
 - C. absolute advantage
 - D. comparable advantage
37. Assume a_{LW} and a_{LC} are the unit labor requirements in wine and cheese production, respectively. The opportunity cost of cheese in terms of wine is
- A. a_{LW}/a_{LC}
 - B. $1/a_{LC}$
 - C. a_{LC}/a_{LW}
 - D. $1/a_{LW}$
38. A country engaging in trade according to the principles of comparative advantage gains from trade because it
- A. is producing exports indirectly more efficiently than it could alternatively.
 - B. is producing imports indirectly using fewer labor units.
 - C. is producing imports indirectly more efficiently than it could domestically.
 - D. is producing exports using fewer labor units.
 - E. is producing exports while outsourcing services.
39. Which of the following statements is TRUE?
- A. A country's exports are determined by its relative high productivity in the industry compared with relative productivity in other sectors.
 - B. A country's exports are determined by its high productivity in the industry compared with that of foreigners.
 - C. A country's exports are determined by its ability to match other countries' productivity.
 - D. A country's exports are determined by its low productivity in the industry compared with that of foreigners.

40. The country of Rhozundia is blessed with rich copper deposits. The cost of copper produced (relative to the cost of widgets produced) is therefore very low. From this information we know that
- A. Rhozundia has a comparative advantage in copper.
 - B. Rhozundia may or may not have a comparative advantage in copper.
 - C. Rhozundia should export both widgets and copper.
 - D. Rhozundia should invest in more widget production.
 - E. Rhozundia should import copper and export widgets.
41. When a country can produce a good or service at a lower opportunity cost than another country, the country is said to have a(n)
- A. absolute advantage
 - B. comparative advantage
 - C. cost advantage
 - D. trading advantage
42. Which of the following statements is TRUE?
- A. Migrating workers often have very similar characteristics as native workers in the receiving country.
 - B. Migrating workers often have very different characteristics than native workers in the receiving country.
 - C. Migrating workers often underperform native workers in the receiving country.
 - D. Migrating workers often outperform native workers in the receiving country.
43. The slope of a country's production possibility frontier is equal to ____ and the optimal production point is located where the slope is equal to _____. Assume that output of good Y is measured on the vertical axis, output of good X is measured on the horizontal axis, MPL is the marginal product of labor with a subscript indicating which good, P is the price of a good, and w is the wage rate.
- A. $-P_X/P_Y$; $-MPL_Y/MPL_X$
 - B. $-MPL_Y/MPL_X$; $-P_X/P_Y$
 - C. $-MPL_Y/w$; $-MPL_F/w$
 - D. $-P_X/w$; $-P_Y/w$
 - E. $-MPL_X/MPL_Y$; $-P_X/P_Y$

44. Refer to the figure above, which shows a country's possible production possibility frontiers and indifference curves. If the country is producing at ____, then moving to ____ will cause utility to ____.
- A. point a; point b; increase
 - B. point c; point b; increase
 - C. point a; point c; remain unchanged
 - D. point c; point b; decrease
 - E. point c; point b; remain unchanged
45. Given the information in the table above, Foreign's opportunity cost of cloth is
- A. 0.5
 - B. 2.0
 - C. 6.0
 - D. 1.5
 - E. 3.0

Problem 1

China and Mexico can both produce drones. Average cost in each country depends on domestic output because of external economies of scale:

$$AC_{China} = 40 - 0.02Q_{China}, \quad AC_{Mexico} = 42 - 0.01Q_{Mexico},$$

where AC_i is the average cost in country i and Q_i is domestic output.

World demand for drones is:

$$Q^{World} = 300.$$

Initially, China produces $Q_{China} = 250$ and Mexico produces $Q_{Mexico} = 50$.

- (1) Compute the average cost in each country and identify the lower-cost producer.
- (2) Suppose Mexico's domestic market expands, raising its drone production to $Q_{Mexico} = 200$. Recompute the average costs for China and Mexico.
- (3) Based on these revised costs, determine which country is more likely to become the world's drone production hub.
- (4) Explain how this example illustrates:
 - path dependence,
 - multiple possible long-run equilibria,
 - the role of expectations in determining industry location when external economies of scale are present.

Problem 2

The United States imports bicycles and faces the domestic demand and supply:

$$Q_D = 1,200 - 3P, \quad Q_S = 2P - 100.$$

The world price of bicycles is $P_W = 200$. The United States is a large importer, and foreign export supply is:

$$P^{Foreign} = 150 + 0.25Q^{Foreign}.$$

The U.S. imposes an import quota of 120 bicycles.

- (1) Find the free-trade domestic price in the U.S. and the free-trade quantity of imports.
- (2) Under the quota, determine:
 - the domestic price P_Q ,
 - the foreign export price $P^{Foreign}$,
 - the value of quota rents.
- (3) Suppose quota licenses are given to U.S. firms. Compute:
 - the production distortion,
 - the consumption distortion,
 - quota rents,
 - the terms-of-trade effect.
- (4) Compare these welfare effects with a tariff that restricts imports to the same quantity. Which policy is more efficient, and why?